PV102 Day 4: Module Behavior String Design and Shade

Students connect PV101 series/parallel measurement habits to PV102 design judgment: string voltage, current, controller limits, temperature, and shade-sensitive layout decisions.

PV102 | CTS Day 20 | Instructor Run Sheet
String design follows limits, temperature, and shade reality. | Thursday, August 6, 2026

Core rule

String design follows limits, temperature, and shade reality.

Learning targets

- Predict how series and parallel wiring affect PV string voltage and current.
- Explain why controller input limits control string design.
- Recognize shade as both a production and mismatch problem.
- Use a safe paper/string model to compare design options before hardware work.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	PV module spec sheet samples String cards Calculator Shade cards Controller limit cards Whiteboard Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 3 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.
12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work

		triggers.
1:00-2:15	String Design Paper Lab	Teams compare three paper string options against controller limits and shade constraints.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 5 turns daily energy need into storage, autonomy, and battery-protection decisions.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use module behavior string design and shade to make the next design decision safer and more defensible. The rule is simple: String design follows limits, temperature, and shade reality."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Read the module and controller cards.
- Build three possible string layouts using cards.
- Calculate approximate operating voltage/current and open-circuit voltage.
- Mark shade exposure from Day 2 scenario notes.
- Reject any option that violates a limit or ignores shade evidence.
- Handoff the safest first-pass string layout and its required verification checks.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 5 turns daily energy need into storage, autonomy, and battery-protection decisions.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.